Differential Gene Expression Analysis of Liver Cancer vs Normal Tissue Using TCGA RNA-Seq Data

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Analysis: Independent computational project using public TCGA-LIHC data

1. Background

Hepatocellular carcinoma (HCC) accounts for the great majority of cases of primary liver cancer, which is still one of the top causes of cancer-related deaths globally. According to global burden studies, the absolute number of new cases and deaths from primary liver cancer grew by 59.4% and 55.8%, respectively, between 2000 and 2023. Since 2020, the age-standardized incidence rates that had been dropping have reversed [1]. The etiology with the greatest rate of growth worldwide is metabolic dysfunction-associated steatotic liver disease (MASLD) [1].

Reactivation of fetal markers and loss of normal hepatocyte function are the two key characteristics of the molecular landscape of HCC. The most popular diagnostic biomarkers in clinical practice are serum alpha-fetoprotein (AFP) and glypican-3 (GPC3). However, AFP's sensitivity and specificity for early-stage HCC are limited [3]. In order to enhance early diagnostic performance, recent research has concentrated on integrating GPC3 with AFP and DCP [4].

However, research on the entire collection of transcriptional alterations causing HCC growth is still ongoing, especially with regard to the integrated processes of cell proliferation, metabolic rewiring, and stromal remodeling. **This study aims to answer the following questions: What biological processes are regulated by the genes that exhibit significantly different expression between tumor and normal liver tissue in TCGA-LIHC?** In order to define the transcriptional signature of HCC and ascertain which biological pathways are most impacted, I analyzed RNA-sequencing data from 374 tumor samples and 50 nearby normal tissue samples.

2. Methods

Data source the Cancer Genome Atlas Liver Hepatocellular Carcinoma project (TCGA-LIHC) provided RNA-sequencing count data via the Genomic Data Commons portal using the TCGAbiolinks R tool [5]. The collection included 50 solid tissue normal samples and 374 primary tumor samples, for a total of 424 samples and 60,660 genes. Workflow data from STAR-Counts were utilized.

Analysis of differential expression: 49,319 genes remained for investigation after low-expression genes (total count < 10 across all samples) were eliminated. DESeq2 v1.3x was used to measure the difference in expression between tumor and normal tissue [6]. Using the Benjamini-Hochberg multiple testing

correction in a negative binomial generalized linear model. Genes were deemed statistically significant if their absolute log2 fold change was greater than 1 and their adjusted p-value (padj) was less than 0.05.

***Functional enrichment*:** Gene Ontology (GO) Biological Process enrichment was performed on significantly upregulated genes using `clusterProfiler` and the `org.Hs.eg.db` annotation database [7]. Terms with adjusted $p < 0.05$ were considered significant.

***Visualization*:** EnhancedVolcano, pheatmap, and clusterProfiler were used to create the volcano plot, heatmap (of the top 50 most significant genes), and GO enrichment dot plot, respectively.

***Software*:** R v4.x was used in RStudio for all analyzes. The full code and outcomes can be found at [https://github.com/salehanayab39].

3. Results

***Expression that varies*:** 9,457 of the 49,319 genes examined showed substantially altered expression between tumor and normal tissue (padj < 0.05, |log2FC| > 1). In tumor tissue, 7,529 (79.6%) of these were upregulated and 1,928 (20.4%) were downregulated, suggesting that the transcriptional pathway in HCC is primarily activating.

***Top genes with varying expression*:** Cell cycle and chromosomal segregation functions were highly enriched among the most markedly elevated genes. By corrected p-value, the top ten elevated genes were:

Gene	log2FC	padj	Function
AFP	+6.00	2.2×10^{-39}	Fetal marker / diagnostic biomarker
GPC3	+5.92	3.5×10^{-80}	Cell surface proteoglycan / HCC marker
ZIC2	+6.70	4.9×10^{-86}	Transcription factor / stemness
CDKN3	+3.90	1.6×10^{-90}	Cell cycle phosphatase
CDC25C	+4.35	5.6×10^{-89}	Mitotic regulator
BIRC5	+4.10	1.3×10^{-82}	Anti-apoptosis (Survivin)
UBE2T	+3.20	3.9×10^{-88}	Ubiquitin-conjugating enzyme
CENPF	+3.88	4.0×10^{-87}	Centromere protein
NUF2	+4.20	4.0×10^{-87}	Kinetochore protein
KIF4A	+3.91	1.3×10^{-82}	Kinesin motor protein

The biological validity of the method was confirmed when AFP and GPC3, two recognized HCC diagnostic indicators, were independently rediscovered by the study. Additionally, SPP1 (+4.30, padj = 6.9×10^{-31}), COL1A1 (+1.80, padj = 1.7×10^{-11}), ALB (−1.84, padj = 2.1×10^{-11}), and APOA1 (−1.33, padj = 2.4×10^{-11}) were found to be additional classic HCC indicators.

***Visualization*:** All 49,319 genes are shown in the volcano plot (Figure 1), with considerably elevated genes in the upper right and downregulated genes in the upper left. Two unique sample groups can be seen in the heatmap of the top 50 genes (Figure 2), which clearly distinguishes between tumor and normal samples. The volcano plot (Figure 1) displays all 49,319 genes, with significantly upregulated genes in the upper right and downregulated genes in the upper left. The heatmap of the top 50 genes (Figure 2) shows clear separation between tumour and normal samples, with two distinct sample clusters visible.

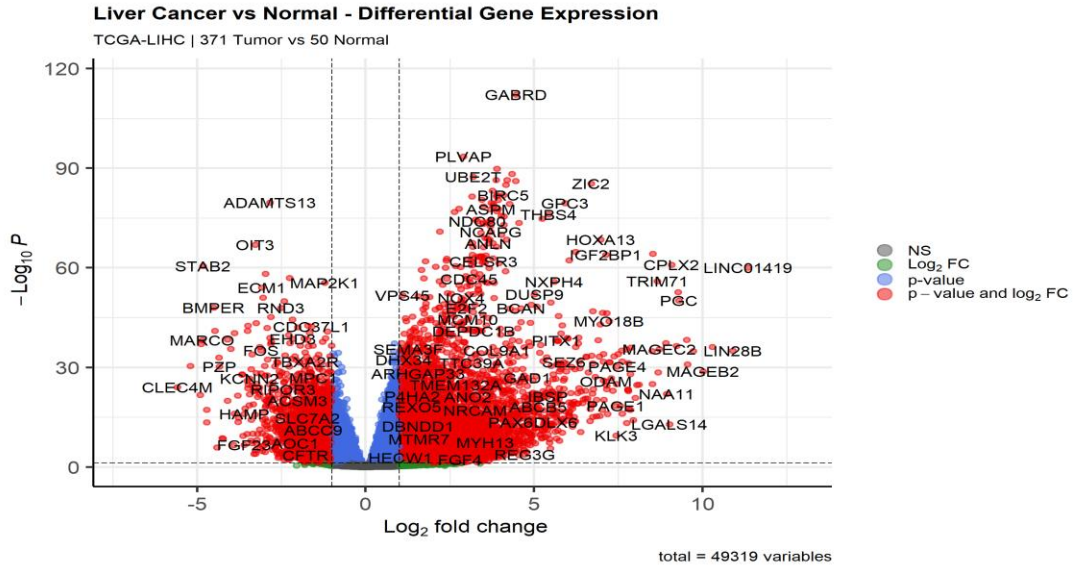


Figure.1 *A volcano diagram illustrating the differences in gene expression between normal and tumor liver tissue. Log2 fold change on the X-axis. •log10 adjusted p-value on the Y-axis. Red dots indicate genes that are significantly elevated (padj < 0.05, log2FC > 1). Blue dots represent genes that are markedly downregulated. Non-significant: gray dots.*

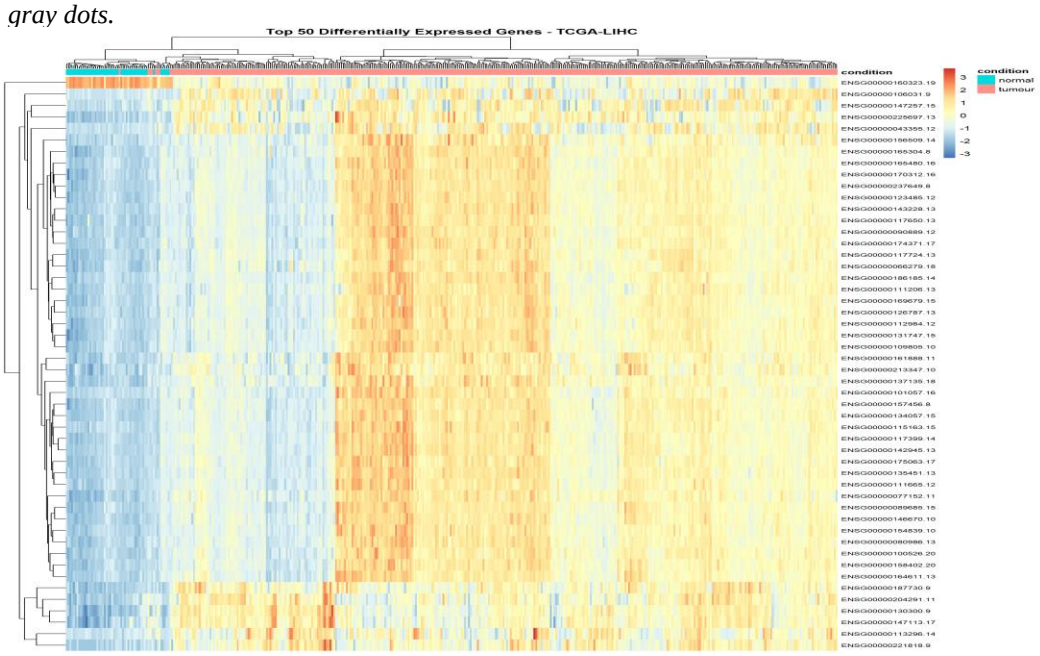


Figure. 2 *Shows a heatmap of the top 50 genes with the greatest degree of differential expression. Genes are rows. Samples are in columns. High expressiveness is shown by red. Blue: expressionless. Tumor and normal tissue are represented by the two separate sample clusters.*

***Gene Ontology enrichment*:** Significant enrichment in cell cycle activities (mitotic nuclear division, chromosomal segregation, DNA replication), microtubule-based motility, and spindle organization was found by GO Biological Process enrichment of upregulated genes. The unchecked cell division that characterizes HCC is consistent with this proliferation profile.

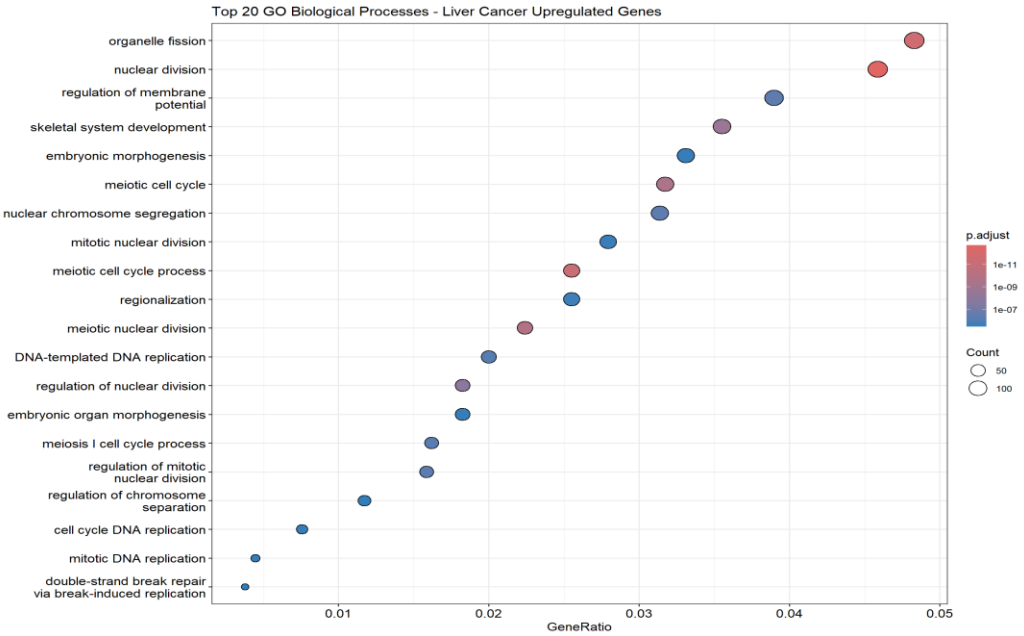


Figure.3 *Enrichment dot plot for upregulated genes in Gene Ontology Biological Process. Dot size: the pathway's number of genes. Dot color: p-value adjustment. The proliferative aspect of HCC is reflected in the prevalence of cell cycle terminology.*

4. Discussion

Using only publicly available data and open-source technologies, this research effectively summarized the known molecular signature of hepatocellular carcinoma. The findings show the loss of mature hepatocyte function (ALB, APOA1 downregulation) as well as the reactivation of fetal/proliferative programs (AFP, GPC3, CDKN3, BIRC5).

***Alignment with published biology*:** The most well-known molecular marker of HCC is the elevation of GPC3 and AFP, which is employed in clinical settings for monitoring and diagnosis [3, 4]. The high rate of HCC cell proliferation is reflected in the strong enrichment of cell cycle genes (CDKN3, CDC25C, BIRC5, KIF4A, CENPF, NUF2). It has been demonstrated that CDKN3 stimulates HCC cell proliferation by increasing G1-S transition, and BIRC5 is at the center of its co-expression network [6]. The loss of hepatocyte differentiation, a well-established characteristic of tumor growth, is indicated by the downregulation of ALB and APOA1, the genes encoding the most prevalent liver-secreted proteins [7].

***Unexpected findings*:** The transcription factor with the strongest upregulation was ZIC2 (log2FC = +6.70). It has been demonstrated that ZIC2 is abundantly expressed in liver cancer stem cells and is necessary for their self-renewal. By attracting the NURF complex to its promoter, ZIC2 activates OCT4 [5]. Given its presence in this dataset, it could be worthwhile to look at it further as a possible therapeutic target.

***Limitations*:** There are a few restrictions to be aware of. First, there is an imbalance in sample numbers (374 tumor vs. 50 normal), which could have an impact on the normal group's statistical power. Second, because samples were processed across several institutions, platforms, and time points, potentially generating technical variance, batch effects are widespread in TCGA data [1]. Third, post-transcriptional regulation means that changes in mRNA do not always correspond to changes in proteins; RNA-seq measures transcript abundance rather than protein levels. Fourth, because the analysis is cross-sectional, causality cannot be established.

***Future directions*:**
The following are logical next steps: (1) survival analysis to determine whether high expression of the top upregulated genes (e.g., ZIC2, BIRC5) predicts poor patient outcome; (2) protein-protein interaction network analysis to identify hub genes and possible therapeutic targets; (3) comparison with other cancer types to determine which changes are pan-cancer versus HCC-specific; and (4) validation in independent cohorts like ICGC or GEO datasets.

References

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